

NMRK and double-spacelike-collinear limits: kinematics and hidden regions on equal footing

Draft note for internal discussion

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1 Purpose and conventions

This note compares two six-point asymptotic expansions using the same graphs, the same external labels and the same final HRF certification criterion: central next-to-multi-Regge kinematics (NMRK), with particular attention to the surface $w_M = z_M$, and the generic double-spacelike-collinear (DSC) limit of Ref. [4]. Neither limit is defined by dual conformal invariance. Dual conformal invariance only reduces the variables on which a particular observable can depend.

The presentation is deliberately not chronological. We first define the two limits symmetrically, then work through the common one-loop hexagon, and only afterwards report and compare the other graphs. Discovery-stage vectors are kept distinct from certified region vectors.

All momenta are outgoing in algebraic formulae. The physical sheet is a $2 \rightarrow 4$ process with particles 1 and 2 incoming. Hence

$$s_{12} > 0, \quad s_{34}, s_{45}, s_{56} > 0, \quad s_{16}, s_{23} < 0, \quad (1)$$

where s_{16} and s_{23} become small in the DSC limit. A region vector always means

$$x_i \longrightarrow \eta^{v_i} x_i, \quad (v_0, \dots, v_{N-1}; 1). \quad (2)$$

Here η denotes the small parameter of the expansion under discussion: the NMRK parameter in Sec. 2.2 or ε_D in Sec. 2.3. The final component is displayed explicitly whenever there is a possible normalization ambiguity.

2 Kinematics: coordinates first, limits second

Two notationally different objects were conflated in an earlier version of this note: a coordinate system for generic six-point kinematics and a limit taken inside that coordinate system. They are separated here. Subscripts M and D distinguish the minimal-light-cone and Duhr–Venkata–Zhang transverse variables.

2.1 General minimal light-cone variables

The minimal light-cone variables (MSLCV) follow Refs. [1, 2]. Set $P = p_4^+$ and $Q = q_1^\perp \bar{q}_1^\perp$. The longitudinal coordinates are defined by

$$p_3^+ = PX_{34}, \quad p_4^+ = P, \quad p_5^+ = \frac{P}{X_{45}}, \quad p_6^+ = \frac{P}{X_{45}X_{56}}, \quad (3)$$

and the transverse coordinates by

$$\begin{aligned} p_4^\perp &= -\frac{q_1^\perp}{z_M - 1}, & p_5^\perp &= \frac{z_M q_1^\perp}{w_M(z_M - 1)}, \\ p_6^\perp &= \frac{z_M(w_M - 1)q_1^\perp}{w_M(z_M - 1)}, & p_3^\perp &= -p_4^\perp - p_5^\perp - p_6^\perp, \end{aligned} \quad (4)$$

with barred equations defining $\bar{z}_M, \bar{w}_M$. In complexified kinematics the barred and unbarred coordinates may be manipulated as independent algebraic variables. On the real Lorentzian $2 \rightarrow 4$ sheet specified in Eq. (1), however, the barred transverse components are the complex conjugates of the unbarred ones. Consequently,

$$\bar{z}_M = z_M^*, \quad \bar{w}_M = w_M^*. \quad (5)$$

Thus the bars on the MSLCV variables used in the NMRK analysis below are literal complex conjugation; algebraic independence is only a device for complexification and analytic continuation. Equations (3)–(4) are a parametrization of generic kinematics; no Regge or collinear limit has yet been taken.

The same coordinates can be reconstructed from invariants. In the present ordering,

$$X_{34} = \frac{s_{13}}{s_{14}}, \quad X_{45} = \frac{s_{14}}{s_{15}}, \quad X_{56} = \frac{s_{15}}{s_{16}}, \quad Q = -\frac{s_{23}X_{34}X_{45}X_{56}}{1 + X_{56} + X_{45}X_{56} + X_{34}X_{45}X_{56}}. \quad (6)$$

For completeness, put $u = z_M - 1$, $\bar{u} = \bar{z}_M - 1$, $v = w_M - 1$, and $\bar{v} = \bar{w}_M - 1$. The transverse coordinates are fixed, up to exchanging the two roots, by

$$\begin{aligned} u\bar{u} &= \frac{s_{23}X_{34}}{s_{24}}, & u + \bar{u} &= \frac{X_{34}^2 + u\bar{u} - s_{34}X_{34}u\bar{u}/Q}{X_{34}}, \\ w_M\bar{w}_M &= \frac{X_{45}(1 + u + \bar{u} + u\bar{u})s_{24}}{s_{25}}, & v\bar{v} &= \frac{s_{26}}{X_{56}s_{25}}, \quad v + \bar{v} = w_M\bar{w}_M - 1 - v\bar{v}. \end{aligned} \quad (7)$$

Equations (6) and (7) will provide the exact bridge from the twistor parametrization below; they are valid before either limit is taken.

2.2 Central NMRK and the additional $w_M = z_M$ surface

Central NMRK is the scaling

$$X_{34} = \frac{\hat{X}_{34}}{\eta}, \quad X_{56} = \frac{\hat{X}_{56}}{\eta}, \quad X_{45}, Q, z_M, \bar{z}_M, w_M, \bar{w}_M = O(1), \quad \eta \rightarrow 0. \quad (8)$$

The positive HRF chart used below is

$$Q, \hat{X}_{34}, X_{45}, \hat{X}_{56}, K_z > 0, \quad K_z = (1 - z_M)(1 - \bar{z}_M) > 0. \quad (9)$$

The condition $w_M = z_M$, $\bar{w}_M = \bar{z}_M$, is a further restriction, not part of the definition of NMRK. For the planar ordering relevant to the central-emission variables,

$$1 - u_A = \frac{X_{45}(w_M - z_M)(\bar{w}_M - \bar{z}_M)}{(1 + X_{45})(w_M\bar{w}_M + X_{45}z_M\bar{z}_M)}. \quad (10)$$

Thus the extra surface is the distinguished cross-ratio boundary $u_A = 1$.

2.3 DSC momentum twistors, labels and physical sheet

We now define the variables actually used by the DSC runner. Hats denote the native cyclic labels of Ref. [4]. A convenient projective gauge for the four undeformed twistors is

$$Z_1 = (1, 0, 0, 0), \quad Z_4 = (0, 1, 0, 0), \quad Z_5 = (1, 1, 1, 0), \quad Z_6 = (1, a, 0, 1), \quad (11)$$

where a is the remaining hard seed modulus. With $\varepsilon_D \rightarrow 0$, the double-collinear deformation is

$$\begin{aligned} Z_2 &= Z_1 + \varepsilon_D Z_4 - \varepsilon_D \tau_1 \frac{\langle 14 \rangle}{\langle 61 \rangle} Z_6 - \varepsilon_D^2 (1 - z_D) \tau_2 \frac{\langle 14 \rangle}{\langle 45 \rangle} Z_5, \\ Z_3 &= Z_4 + \varepsilon_D Z_1 - \varepsilon_D \tau_2 \frac{\langle 14 \rangle}{\langle 45 \rangle} Z_5 + \varepsilon_D^2 \frac{(1 - \bar{z}_D) \tau_1}{\bar{z}_D} \frac{\langle 14 \rangle}{\langle 61 \rangle} Z_6. \end{aligned} \quad (12)$$

The paper's momentum fractions obey $\tau_i = (1 - \xi_i)/\xi_i$. The variables $z_D, \bar{z}_D$ in Eq. (12) are not the MSLCV variables $z_M, \bar{z}_M$, despite the deliberately similar notation.

It is important to state the reality condition before continuing to the DSC sheet. In the standard real transverse-momentum realization of the six-point $2 \rightarrow 4$ multi-Regge region, the variables denoted by z_D and $\bar{z}_D$ form a complex-conjugate pair,

$$\bar{z}_D = z_D^*. \quad (13)$$

This is the branch on which the familiar MRK formulae are written in terms of $|z_D|$ and $z_D/\bar{z}_D$. Duhr–Venkata–Zhang reach the double-spacelike-collinear region by analytic continuation from the timelike Euclidean region into a specified Mandelstam region. During this continuation z_D and $\bar{z}_D$ must be regarded as the two algebraic roots, not as a conjugate pair at every intermediate point.

The change of reality is transparent from their cross-ratio variables. If

$$r_1 = \frac{u_1}{1 - u_2}, \quad r_3 = \frac{u_3}{1 - u_2}, \quad (14)$$

then their defining relations imply

$$z_D \bar{z}_D = \frac{r_1}{r_3}, \quad z_D + \bar{z}_D = 1 + \frac{r_1 - 1}{r_3}. \quad (15)$$

Thus z_D and $\bar{z}_D$ are the roots of a quadratic with real coefficients whenever the continued ratios are real. A negative quadratic discriminant gives Eq. (13); after the discriminant changes sign the two roots are both real. The latter is the branch used by the DSC HRF chart below.

For clarity, the twistor-to-invariant operation used below is also fixed here. Write $Z_i = (\lambda_i, \mu_i)$, take all labels cyclically, and set

$$\tilde{\lambda}_i = \frac{\langle i \ i+1 \rangle \mu_{i-1} + \langle i+1 \ i-1 \rangle \mu_i + \langle i-1 \ i \rangle \mu_{i+1}}{\langle i-1 \ i \rangle \langle i \ i+1 \rangle}, \quad s_{ij}^{(T)} = \langle ij \rangle [ji]. \quad (16)$$

Angle and square brackets are the two-by-two determinants of the corresponding spinors. Equations (11)–(16) therefore determine every pair invariant rationally at finite ε_D .

The native twistor labels are not the graph labels. The order-aware map used in the runner is

$$p_1 = \hat{p}_1, \quad p_2 = \hat{p}_3, \quad p_3 = \hat{p}_4, \quad p_4 = \hat{p}_5, \quad p_5 = \hat{p}_6, \quad p_6 = \hat{p}_2. \quad (17)$$

It maps the two small twistor pairs to the desired graph invariants s_{23} and s_{16} . To keep the sign convention transparent, let $s_{ij}^{(T)} = \langle ij \rangle [ji]$ denote the invariants reconstructed from the twistors in the seed convention above. Their leading adjacent values are

$$\begin{aligned} s_{12}^{(T)} &= \frac{\tau_2}{a(1+\tau_1)(1+\tau_2)} + O(\varepsilon_D), & s_{23}^{(T)} &= -\frac{\varepsilon_D^2 \tau_1}{a \bar{z}_D} + O(\varepsilon_D^3), \\ s_{34}^{(T)} &= -\frac{1}{(a-1)(1+\tau_2)} + O(\varepsilon_D), & s_{45}^{(T)} &= \frac{1}{a}, \\ s_{56}^{(T)} &= -\frac{\tau_1}{(a-1)(1+\tau_1)} + O(\varepsilon_D), & s_{16}^{(T)} &= -\frac{\varepsilon_D^2 \tau_2 z_D}{a} + O(\varepsilon_D^3). \end{aligned} \quad (18)$$

Our graph-polynomial convention is

$$s_{ij} = -s_{ij}^{(T)}. \quad (19)$$

This is a single spinor-bracket/metric convention, not a separate sign choice for each invariant.

A convenient *real slice* of the physical $2 \rightarrow 4$ sheet is

$$a < 0, \quad -1 < \tau_1 < 0, \quad \tau_2 < -1, \quad z_D < 0, \quad \bar{z}_D < 0. \quad (20)$$

On this slice $(s_{12}^{(T)}, s_{23}^{(T)}, s_{34}^{(T)}, s_{45}^{(T)}, s_{56}^{(T)}, s_{16}^{(T)})$ has signs $(-, +, -, -, -, +)$. Equation (19) therefore gives exactly Eq. (1). The bar on $\bar{z}_D$ is inherited notation from the conjugate branch in Eq. (13). On the continued branch (20), however, z_D and $\bar{z}_D$ are two real roots and need not be equal. Requiring both roots to be negative selects the sign-definite component used by HRF; it does not reduce the kinematics to the diagonal $z_D = \bar{z}_D$. This is different from Eq. (5): the MSLCV variables remain complex conjugates on the physical NMRK sheet considered in this note.

2.4 Exact coordinate map and comparison of the two limiting paths

There is no additional informal “dictionary.” At finite ε_D , reconstruct the fifteen s_{ij} using Eqs. (11)–(19), and insert them in the invariant relations (6) and (7). This is an exact, invariantly defined map from $(a, \tau_1, \tau_2, z_D, \bar{z}_D, \varepsilon_D)$ to the MSLCV.

Expanding that exact map makes the limiting path explicit:

$$\begin{aligned} X_{34} &= \frac{a-1}{1+\tau_2} + O(\varepsilon_D), & X_{45} &= -\frac{1}{a} + O(\varepsilon_D), \\ X_{56} &= \frac{a}{(a-1)(1+\tau_1)\tau_2 z_D} \varepsilon_D^{-2} + O(\varepsilon_D^{-1}), & Q &= \frac{\tau_1}{a\tau_2 \bar{z}_D} \varepsilon_D^2 + O(\varepsilon_D^3). \end{aligned} \quad (21)$$

The approach of the transverse variables to the boundary is fixed by

$$\begin{aligned} u\bar{u} &= \frac{(a-1)^2 \tau_1}{a\tau_2 \bar{z}_D} \varepsilon_D^2 + O(\varepsilon_D^3), & u + \bar{u} &= -\frac{(a-1)(\tau_1 + a\tau_2 \bar{z}_D)}{a\tau_2 \bar{z}_D} \varepsilon_D + O(\varepsilon_D^2), \\ v\bar{v} &= \frac{(a-1)^2 \tau_1 \tau_2 z_D}{a} \varepsilon_D^2 + O(\varepsilon_D^3), & v + \bar{v} &= \frac{(a-1)(\tau_1 + a\tau_2 z_D)}{a} \varepsilon_D + O(\varepsilon_D^2). \end{aligned} \quad (22)$$

Thus DSC has $X_{34}, X_{45} = O(1)$, $X_{56} = O(\varepsilon_D^{-2})$, $Q = O(\varepsilon_D^2)$, and $z_M, \bar{z}_M, w_M, \bar{w}_M \rightarrow 1$. Central NMRK instead has both X_{34} and X_{56} large while Q remains hard. Generic DSC and central NMRK with $w_M = z_M$ are therefore distinct paths in the same six-point kinematic space. Their one-loop cancellation hypersurfaces are nevertheless related, as shown explicitly in Sec. 3.

3 Common one-loop hexagon

The graph and all label conventions are displayed in Fig. 1. The loop edges

$$(5, 6), (6, 1), (1, 2), (2, 3), (3, 4), (4, 5) \longleftrightarrow x_0, x_1, x_2, x_3, x_4, x_5, \quad (23)$$

respectively. The external attachments are $p_1 \rightarrow 1$, $p_2 \rightarrow 4$, $p_3 \rightarrow 5$, $p_4 \rightarrow 3$, $p_5 \rightarrow 2$, and $p_6 \rightarrow 6$.

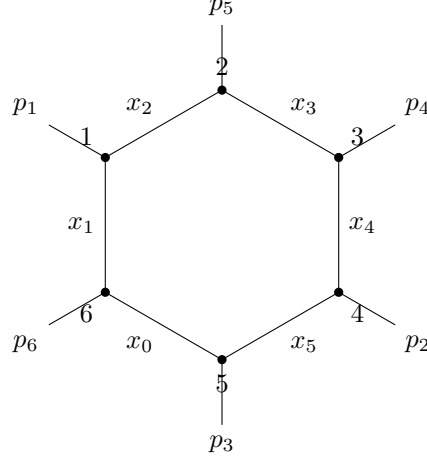


Figure 1: The one-loop hexagon, with the graph vertices, external momenta and Schwinger parameters used by HRF.

Before either kinematic expansion, the Symanzik polynomials are

$$\mathcal{U} = x_0 + x_1 + x_2 + x_3 + x_4 + x_5. \quad (24)$$

and

$$\begin{aligned} \mathcal{F} = & s_{16}x_0x_2 + (-s_{14} + s_{23} - s_{45} - s_{46})x_0x_3 + s_{23}x_0x_4 \\ & + (-s_{14} - s_{16} + s_{23} - s_{45} - s_{46} - s_{56})x_1x_3 \\ & + (-s_{16} + s_{23} - s_{46} - s_{56})x_1x_4 \\ & + (s_{12} - s_{34} - s_{35} - s_{45} - s_{46} - s_{56})x_1x_5 \\ & + s_{45}x_2x_4 + (s_{16} - s_{23} - s_{34} - s_{35})x_2x_5 \\ & + (-s_{14} - s_{34} - s_{45} - s_{46})x_3x_5. \end{aligned} \quad (25)$$

The same $\mathcal{U}$, $\mathcal{F}$, graph labels and HRF implementation are used for both expansions. Only the substitution for the invariants s_{ij} changes.

3.1 Certification rule after asymptotic-order alignment

This subsection fixes the notation used in the two examples. Write the full kinematically expanded polynomial as

$$\mathcal{F}(\eta, \mathbf{x}) = \sum_m \eta^{a_m} c_m \mathbf{x}^{\mathbf{r}_m}. \quad (26)$$

The algorithm has three logically separate stages.

1. Asymptotic-order alignment (face discovery). A trial face vector f assigns the total weight $a_m + \mathbf{r}_m \cdot f$ to each monomial. The monomials of minimum weight define an order-aligned face $\mathcal{F}^{(f)}$. This operation exposes monomials that occur at different fixed- $\mathbf{x}$ orders in η , but acquire a common effective order after the parameter scaling $x_e \mapsto \eta^{f_e} x_e$, so that they can participate in the same cancellation sector. Because $\mathcal{F}$ is homogeneous in the Schwinger parameters, replacing $f \rightarrow f + c(1, \dots, 1)$ does not change which monomials lie on the face. Thus f is only a discovery-stage representative, not yet a physical region vector.

2. Ordinary HRF on the selected face. Once $\mathcal{F}^{(f)}$ has been selected, the usual HRF analysis is performed just as it would be without asymptotic-order alignment. HRF finds non-monomial cancellation factors $L_i(\mathbf{x})$, forms their ideal $I = \langle L_1, \dots, L_r \rangle$, and decomposes

$$\mathcal{F}^{(f)} = \mathcal{F}_{\text{SL}} + \mathcal{F}_{\text{obs}}, \quad \mathcal{F}_{\text{SL}} \in I. \quad (27)$$

It then finds a cancellation scaling h for which all monomials of $\mathcal{F}_{\text{SL}}$ have the common weight W_{SL} , while the first surviving terms of $\mathcal{U} + \mathcal{F}_{\text{obs}}$ have weight $W_{\text{HR}} > W_{\text{SL}}$. We call h the *HRF face scaling*. The displayed sum $f + h$ is useful during discovery, but it inherits the uniform ambiguity of f and is not automatically the final vector.

3. Certification in the original coordinates. The algorithm returns to the full $\mathcal{P}(\eta, \mathbf{x}) = \mathcal{U} + \mathcal{F}$ and retains every occupied weight layer. Each $\mathcal{F}$ -layer is resolved in the filtration I^q/I^{q+1} . A layer in I^q cancels to order q at the common pinch; an absent power of η is not counted as a cancellation layer. The first resolved nonvanishing layer fixes W_{HR} . The simultaneous balance with $\mathcal{U}$ fixes the previously ambiguous uniform component and, with the convention in Eq. (2), produces the final original-coordinate vector v .

If the resolved W_{HR} monomials already define a unique lower facet of the full Lee–Pomeransky polynomial, this gives a direct certificate. If the ideal-layer system is nonlinear, singular or non-unique, a local dissection provides the necessary and sufficient facet test and its pullback fixes v . No space-time dimensional information enters any of these decisions.

3.2 NMRK with $w_M = z_M$

The cancellation surface factorizes as

$$\begin{aligned} L_1^{\text{NMRK}} &= (1 + X_{45})x_4 - \hat{X}_{34}X_{45}x_5, \\ L_2^{\text{NMRK}} &= (1 + X_{45})x_2 - K_z X_{45} \hat{X}_{56}x_1. \end{aligned} \quad (28)$$

Up to a nonzero kinematic factor,

$$\mathcal{F}_{\text{SL}}^{\text{NMRK}} = Q \hat{X}_{34} \hat{X}_{56} L_1^{\text{NMRK}} L_2^{\text{NMRK}}. \quad (29)$$

For generic NMRK the corresponding bilinear coefficient determinant is proportional to $(w_M - z_M)(\bar{w}_M - \bar{z}_M)$; it is a near miss. The surface $w_M = z_M$ makes the factorization exact.

For this case the discovery and certification data are

$$\begin{aligned} f &= (-1, -1, -2, -1, -2, -1), \\ h &= (-1, -1, -1, -1, -1, -1), \\ v_{\text{NMRK}} &= f + h = (-2, -2, -3, -2, -3, -2), \quad (v_{\text{NMRK}}; 1), \end{aligned} \quad (30)$$

with

$$W_{\text{SL}} = -4, \quad W_{\text{HR}} = -3, \quad W_{\text{HR}} - W_{\text{SL}} = 1. \quad (31)$$

Here the direct composition is already the unique lower-facet normal. At W_{HR} , the active $\mathcal{U}$ contribution is $x_2 + x_4$, and the active $\mathcal{F}$ contribution is

$$-Q\hat{X}_{34}X_{45}\hat{X}_{56}(K_zx_0x_2 + K_zx_0x_4 + \hat{X}_{34}x_3x_5 + K_z\hat{X}_{56}x_1x_3). \quad (32)$$

These monomials, together with the resolved promoted face, certify a scaleful region.

3.3 Generic DSC

The DSC cancellation factors are

$$L_1^{\text{DSC}} = (1 + \tau_2)x_4 + x_5, \quad L_2^{\text{DSC}} = \tau_1x_1 + (1 + \tau_1)x_2, \quad (33)$$

and

$$\mathcal{F}_{\text{SL}}^{\text{DSC}} = -(a - 1)a^2\bar{z}_D^2L_1^{\text{DSC}}L_2^{\text{DSC}}. \quad (34)$$

These factors make the contrast with the crossed double-timelike-collinear (DTC) domain immediate. In the DSC domain $-1 < \tau_1 < 0$ and $\tau_2 < -1$, each factor has terms of opposite sign and admits a zero with positive Feynman parameters:

$$L_1^{\text{DSC}} = 0 : \quad x_5 = -(1 + \tau_2)x_4 > 0, \quad L_2^{\text{DSC}} = 0 : \quad x_2 = -\frac{\tau_1}{1 + \tau_1}x_1 > 0. \quad (35)$$

The two equations can therefore hold simultaneously in the interior of the integration domain. For two final-state timelike splittings, by contrast, the momentum fractions satisfy $0 < \xi_i < 1$, and hence

$$\tau_i = \frac{1 - \xi_i}{\xi_i} > 0. \quad (36)$$

It follows that both L_1^{DSC} and L_2^{DSC} are strictly positive for positive x_i . Crossing thus preserves the algebraic form of this candidate generator but removes its common positive Landau pinch. This excludes the timelike continuation of the DSC hidden region; by itself it does not exclude a different cancellation generator in the DTC limit.

The discovery-stage vectors are

$$\begin{aligned} f &= (-2, -2, -2, -1, -2, -2), \\ h &= (-1, -1, -1, -1, -1, -1), \\ f + h &= (-3, -3, -3, -2, -3, -3). \end{aligned} \quad (37)$$

The last line is not the final region vector. The full nonzero coefficient layers satisfy

$$\text{ord}_I \mathcal{F}_{-4} = 2, \quad \text{ord}_I \mathcal{F}_{-3} = 1, \quad \text{ord}_I \mathcal{F}_{-2} = 0, \quad I = \langle L_1^{\text{DSC}}, L_2^{\text{DSC}} \rangle. \quad (38)$$

Here $\mathcal{F}_k$ is the sum of monomials of total weight k , and $\text{ord}_I \mathcal{F}_k = q$ means $\mathcal{F}_k \in I^q$ but $\mathcal{F}_k \notin I^{q+1}$. Thus the -3 layer is present but cancels to first order on the common pinch. Solving the ideal-jet balance gives transverse weights $q_1 = q_2 = 1$, and $\mathcal{U}$ fixes the uniform component. The certified answer is

$$v_{\text{DSC}} = (-2, -2, -2, 0, -2, -2), \quad (v_{\text{DSC}}; 1), \quad (39)$$

with

$$W_{\text{SL}} = -4, \quad W_{\text{HR}} = -2, \quad W_{\text{HR}} - W_{\text{SL}} = 2. \quad (40)$$

This is an original-coordinate ideal-jet certificate. Dissection pulls back to the same vector and explains the previously observed factor of two.

3.4 What is common and what is different

After the separate kinematic rescalings have been made, the coefficients in the two factorized $\mathcal{F}_{\text{SL}}$ polynomials are identified by

$$\frac{K_z X_{45} \hat{X}_{56}}{1 + X_{45}} = -\frac{\tau_1}{1 + \tau_1}, \quad \frac{\hat{X}_{34} X_{45}}{1 + X_{45}} = -\frac{1}{1 + \tau_2}. \quad (41)$$

This is a map of the cancellation factors on the two selected faces, not a claim that the full kinematic limits coincide. Consequently the factorized W_{SL} support is the same. The expansions are nevertheless distinct because the next occupied layers differ. NMRK with $w_M = z_M$ has a depth-one hierarchy; generic DSC has an additional first-order cancellation layer and hence a depth-two hierarchy. This is why the two final vectors are related but not equal.

Expansion	certified v	$(W_{\text{SL}}, W_{\text{HR}})$	gap
NMRK with $w_M = z_M$	$(-2, -2, -3, -2, -3, -2)$	$(-4, -3)$	1
generic DSC	$(-2, -2, -2, 0, -2, -2)$	$(-4, -2)$	2

4 Results within each limit

4.1 NMRK results

The one-loop result above and all three two-loop controls have now been passed through the current final audit. “Full” means that no Feynman parameter is fixed to zero; “boundary” denotes a certified scaling vector on a one-edge boundary stratum. At two loops, a certified vector is not yet identified with a distinct physical hidden region: vectors must also be compared at the level of their polynomial generator and local facet. Every vector in the tables below has last component 1. When a vector is called “relative,” its uniform representative is chosen so that $\max_i v_i = 0$; this does not suppress or alter the final component.

Graph	generic NMRK	certified vectors at $w_M = z_M$	support split
one-loop hexagon	no HR	1 certified	1 full
planar hexbox	no staged candidate	11 certified	7 full + 4 boundary
non-planar hexbox	no staged candidate	3 certified	1 full + 2 boundary
hexagon-pentagon	no staged candidate	no staged candidate	—

Planar-hexbox generator multiplicity

A cancellation hypersurface is defined by non-monomial polynomial factors. No individual Feynman parameter x_i is a cancellation hypersurface, and no monomial factor is counted as an HR generator. For the seven full-support planar-hexbox vectors, introduce the non-monomial linear polynomials

$$\begin{aligned} A &= (1 + X_{45})x_3 - K_z X_{45} \hat{X}_{56} x_1, & B &= (1 + X_{45})x_3 - K_z X_{45} \hat{X}_{56} x_2, \\ C &= (1 + X_{45})x_3 - K_z X_{45} \hat{X}_{56} (x_1 + x_2), & D &= (1 + X_{45})x_5 - \hat{X}_{34} X_{45} x_6, \\ E &= (1 + X_{45})x_5 - \hat{X}_{34} X_{45} x_7, & F &= (1 + X_{45})x_5 - \hat{X}_{34} X_{45} (x_6 + x_7). \end{aligned} \quad (42)$$

Every factor in Eq. (42) contains at least two Schwinger monomials. Up to factor ordering and overall signs, the seven vectors form five generator classes,

$$G_1 = AD, \quad G_2 = CD, \quad G_3 = AF, \quad G_4 = AE, \quad G_5 = BD. \quad (43)$$

The detailed assignment is

row	generator	relative certified vector
1	G_1	$(-2, -1, -1, -2, -1, -2, -1, 0, -1)$
2	G_2	$(-1, -1, -1, -2, -1, -2, -1, 0, -2)$
3	G_3	$(-1, -1, 0, -2, -1, -2, -1, -1, -2)$
4	G_1	$(-1, -1, 0, -2, -1, -2, -1, 0, -2)$
5	G_1	$(-1, -1, 0, -2, -1, -2, -1, 0, -1)$
6	G_4	$(-1, -1, 0, -2, -1, -2, 0, -1, -2)$
7	G_5	$(-1, 0, -1, -2, -1, -2, -1, 0, -2)$

Rows 1, 4 and 5 therefore have exactly the same polynomial generator $G_1 = AD$, but different certified vectors. The present lower-facet audit proves that each vector is scaleful; it does not prove that these three vectors define three physically distinct hidden regions. That distinction requires a common local-coordinate analysis around $A = D = 0$, followed by a single dissection/facet enumeration and a comparison of the active monomial sets and normal cones. The count “seven” must consequently be read as seven certified full-support vectors, not seven established physical HRs.

4.2 DSC results

The saved DSC scans have all been passed through the current structural audit. “Staged” counts face presentations before the final scalefulness/layer test; “certified” counts accepted structural vectors.

Graph	staged presentations	certified	certified vector(s)
one-loop hexagon	1 structural class	1	$(-2, -2, -2, 0, -2, -2)$
planar hexbox	156	1	$(0, -2, 0, -2, 0, -2, -2, 0, 0)$
non-planar hexbox	61	0	–
hexagon-pentagon	0	0	–

For the planar hexbox, 27 structural representatives were audited: 26 fail the resolved W_{HR} -facet test and one has an accepted layered common pinch facet. The non-planar hexbox has six structural representatives, all rejected by the resolved facet test.

5 Graph-by-graph comparison

The one-loop hexagon shows the closest relationship. Both limits reach the same factorized singular hypersurface, but DSC contains one extra occupied cancellation layer. This changes the hierarchy gap from 1 to 2 and changes the final original-coordinate vector.

For the planar hexbox, NMRK with $w_M = z_M$ has seven full-support certified vectors and four certified vectors on one-edge boundary strata. All have hierarchy gap 1 and direct resolved-facet certificates. The seven interior vectors represent only five polynomial-generator classes, and three of them share $G_1 = AD$, as detailed above. Generic DSC instead has one full-support certified vector,

$$v_{\text{DSC}}^{\text{P}} = (0, -2, 0, -2, 0, -2, -2, 0, 0), \quad (W_{\text{SL}}, W_{\text{HR}}) = (-4, -2), \quad (44)$$

with gap 2 and a layered common-pinch dissection certificate. None of the seven full-support NMRK relative vectors equals this DSC vector. This proves that the displayed scaling vectors differ, but it

does not by itself count distinct physical regions. A generator-level translation and common local dissection are required before deciding how many planar-hexbox HRs the two limits share.

For the non-planar hexbox, NMRK with $w_M = z_M$ has one full-support certified vector and two certified vectors on one-edge boundary strata, again with gap 1. The full-support relative vector is

$$(-1, -1, -2, -1, -2, -1, -1, 0, -2), \quad (45)$$

whereas generic DSC has no certified region. The hexagon-pentagon is a common negative control: neither expansion has even a staged candidate.

6 Reproducibility

The calculations are separated by expansion:

- `06_NMRK_wz_AsymptoticOrderAlignment_Checks.nb` contains only NMRK inputs and certificates;
- `07_DSC_AsymptoticOrderAlignment_Checks.nb` contains only DSC inputs and certificates;
- `HRF_RunNMRKMultiGraphAsymptoticOrderAlignment.wl` reruns the NMRK two-loop graphs with the current final audit;
- `HRF_RunGenericDSCAsymptoticOrderAlignment.wl` reruns the DSC graphs.

This document is the only layer that translates and compares the two expansions.

References

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- [3] E. Gardi, F. Herzog, S. Jones and Y. Ma, “Dissecting polytopes: Landau singularities and asymptotic expansions in $2 \rightarrow 2$ scattering,” *JHEP* **08** (2024) 127, [arXiv:2407.13738](#).
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